Aanchala Ravindra Bhongade

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Research Profile

MSc Data Science candidate with a single-author peer-reviewed publication and experience designing reproducible machine-learning experiments, processing large scientific datasets, and building scalable cloud systems. Current research investigates length generalisation in Transformer sequence models using PyTorch, controlled synthetic tasks, and failure diagnostics. Experienced in Python, statistical analysis, HPC/SLURM, and AWS infrastructure.

Education

University of Bristol — Sep 2025 – Expected Sep 2026
MSc Data Science — *Bristol, United Kingdom*

- Relevant study: Statistical Computing and Empirical Methods; Programming and Algorithms; AI and Text Analytics; Large-Scale Data Engineering; Data Science Methods and Practice; Visual Analytics.

Veermata Jijabai Technological Institute (VJTI) — Dec 2021 – Jun 2025
B.Tech. Production Engineering, First Class, CGPA: 7.33/10 — *Mumbai, India*

- Relevant study: Probability and Statistics; Operations Research; Business Analytics; Programming and Problem Solving in C++; Supply Chain Management.

Publication

Bhongade, A. (2025). *The Unequal Impact of Lethal Heatwaves in India for Males and Females: Insights from Climate* Projection Data.

- Single-author peer-reviewed article in the *Cambridge Journal of Climate Research*, 2(1), 182–194, using CMIP6 projections, derived climate indicators, scenario comparison, statistical analysis, and scientific visualisation.

Research Experience

University of Bristol — June 2025 – Present
MSc Research Dissertation – Length Generalisation in Transformer Sequence Models — *Bristol, United Kingdom*

- Built a configuration-driven PyTorch pipeline on AWS compute to generate controlled algorithmic tasks, train Transformer baselines on short sequences, and evaluate extrapolation at sequence lengths up to 1,024 using token-level and exact-match metrics.
- Diagnosed that an initial copy benchmark allowed trivial position-wise identity mapping and redesigned it as delayed copy to isolate long-range memory and length extrapolation.
- Achieved 100% exact-match accuracy at the trained length in preliminary addition experiments, followed by a drop to 0.8% at the first unseen length; designing comparisons of learned absolute, sinusoidal, NoPE, RoPE, and ALiBi representations across tasks and random seeds.

University of Cambridge — Sep 2024 – Mar 2025
Visiting Researcher – Cambridge Collective Intelligence and Design Group — *Cambridge, United Kingdom*

- Built a Python pipeline integrating World Bank CMIP6 climate and demographic projections across five SSP scenarios with appliance-sales data to analyse heat exposure and cooling-energy demand in India through 2100.
- Evaluated two genders and 20 five-year age groups across four future periods, generating 40 scenario-specific contour plots alongside heat-index trends, percentile bounds, and Cooling Degree Day correlation analyses.
- Identified a rise in heat index to approximately 89°F by 2099 under SSP5-8.5, with higher cooling demand among older age groups and substantially wider uncertainty under high-emission scenarios. And translated the work into a single-author peer-reviewed publication.

University of Zurich, *Relic-Kappa Project* — Jul 2024 – Jan 2025
Undergraduate Scientific Computing Researcher — *Zurich, Switzerland (Remote)*

- Processed FITS/HEALPix weak-lensing convergence maps on the UZH Science Cluster across six simulations, five redshifts ($z = 0.6$–2.0), and four Newtonian/relativistic configurations, covering 120 configuration-redshift combinations.
- Developed Python workflows using Astropy, Healpy/Anafast, NumPy, and Matplotlib to compute angular power spectra and produce analysis-ready map statistics and visualisations for downstream research.
- Compared Newtonian and relativistic outputs using mean and relative-difference calculations, 20-bin aggregation, standard deviations, and error bars across simulation runs.

Technical Projects

Elastic Event-Driven Data Pipeline on AWS
EC2, S3, SQS, DynamoDB, IAM, CloudWatch, Auto Scaling

- Deployed an event-driven processing pipeline using S3, SQS, Ubuntu EC2 workers, and DynamoDB, with least-privilege IAM roles and queue-depth scaling through CloudWatch alarms.
- Load-tested the architecture with a 120-file burst; it scaled from one to two workers, processed the complete workload without data loss, and returned to minimum capacity after the queue emptied.

Robust Prediction Intervals Under Distribution Shift
Research Proposal

- Designed a controlled evaluation of four probabilistic forecasters with native and conformal calibration under four temporal shift types and three severity levels, using coverage, interval width, CRPS, and Winkler score.

Robust Time-Series Anomaly Detection Under Instrument Shift
Python, Lightkurve, Astropy, scikit-learn (Ongoing)

- Developing a reproducible pipeline for Kepler and TESS light curves using quality-flag filtering, missing-data handling, detrending, normalisation, and rule-based signal screening.
- Building Isolation Forest anomaly scoring for noisy astronomical observations and designing controlled comparisons against robust statistical baselines.
- Designing cross-quarter and cross-instrument evaluation to distinguish rare temporal behaviour from detector-specific distribution shift using precision-recall and false-positive analysis.

Technical Skills

Programming: Python, C++, R, Git, Linux, Jupyter, LaTeX

Machine Learning: PyTorch, TensorFlow, scikit-learn, Transformers, neural networks, regression, clustering, anomaly detection, time-series analysis, model evaluation, distribution shift, uncertainty quantification

Scientific Computing: NumPy, SciPy, pandas, Matplotlib, xarray, netCDF4, Astropy, Lightkurve, Healpy, Anafast, FITS, HEALPix

Cloud and Infrastructure: AWS EC2, S3, SQS, DynamoDB, IAM, CloudWatch, Auto Scaling, AWS CLI; SLURM and HPC workflows

Research Methods: Experimental design, statistical analysis, reproducible pipelines, ablation studies, error analysis, scientific visualisation

Leadership, Awards and Training

- AWS Academy Cloud Foundations; HarvardX R Basics; CBSE Certificate of Merit for ranking in the top 0.1% nationally in Social Science.
- **Department Manager, Enthusia VJTI:** led volunteer logistics for 500+ participants.
- **Public Relations Executive, Technovanza VJTI:** supported engagement for a national technical festival with 60,000+ attendees.
- **University Basketball Team Captain:** Participated in 7 matches and won 3 inter-college matches throughout my captaincy year.